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China sets new speed record on space-ground laser communication

Technology 22:05, 30-Jan-2026
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China has successfully conducted an operational application experiment of space-ground laser communication at a rate exceeding 100 gigabits per second (Gbps), marking a major advancement in the country's high-speed space data transmission capabilities.



The laser communication ground station in Tashkurgan Tajik Autonomous County, Kashi Prefecture, Xinjiang Uygur Autonomous Region, northwest China. /CMG

The experiment achieved a communication speed of 120 Gbps and high-quality remote-sensing images were successfully processed from the transmitted data, the Aerospace Information Research Institute (AIR) under the Chinese Academy of Sciences announced on Friday.

Test results confirmed a stable communication link with high-quality downlink data transmission. This achievement represents another milestone for the research team, following previous breakthroughs of 10 Gbps in 2023 and 60 Gbps in 2025.

This accomplishment not only sets a new domestic record for space-ground laser communication transmission speed but also overcomes the challenges of rapidly establishing and maintaining

long-term stability, and ensuring efficient and reliable transmission of ultra-high-speed space-ground laser links, according to AIR.

"If we compare space-ground laser communication to building a bridge over a turbulent river, achieving 10 Gbps transmission is like constructing a single-lane bridge with a relatively simple structure," said Li Yalin, senior engineer at AIR and technical lead of the team.

"Achieving 120 Gbps transmission is akin to building a multi-lane highway bridge. It not only requires rapid construction to establish connections quickly but also must ensure extremely high traffic efficiency with multiple lanes operating in parallel. The engineering difficulty increases exponentially," said Li.

Li noted that space-ground laser communication has become the optimal solution for achieving ultra-high-speed massive data transmission between satellites and ground stations.

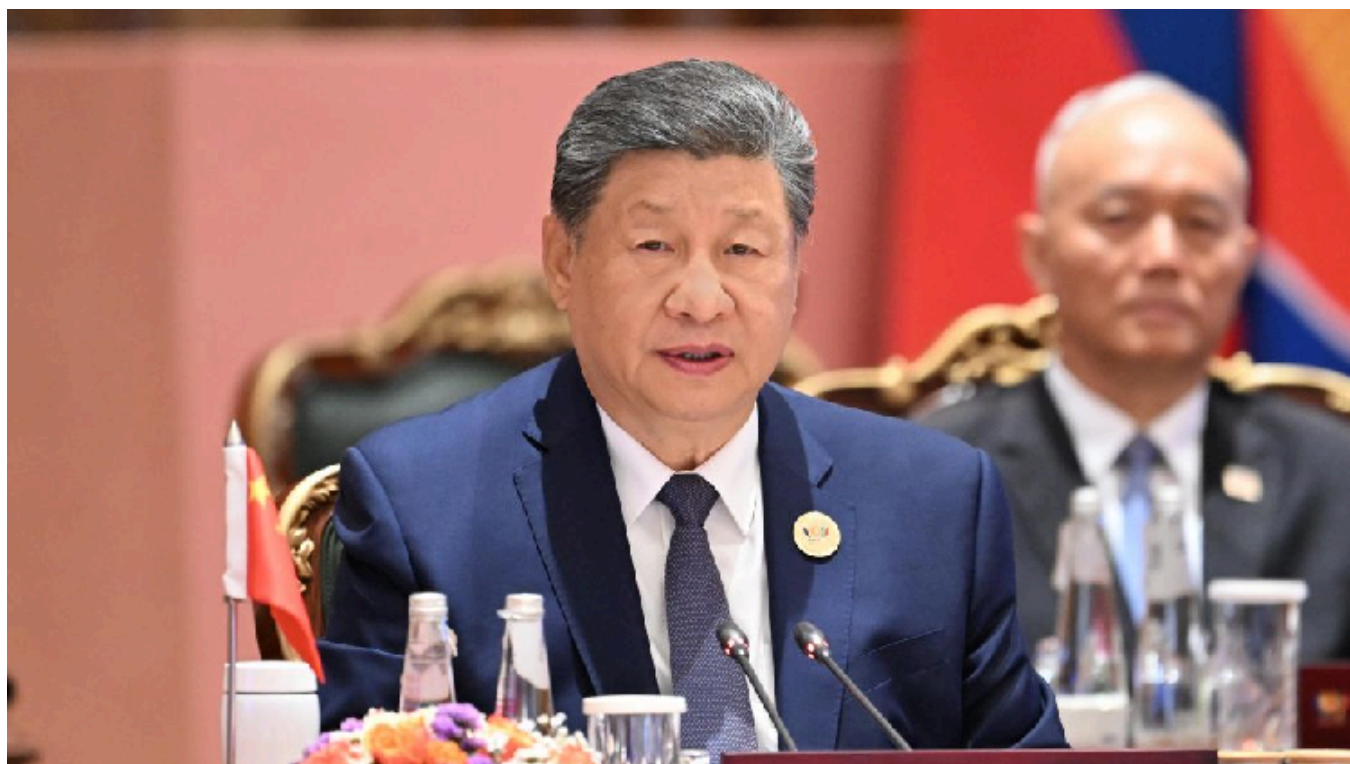
The experiment was jointly conducted using a self-developed 500mm aperture space-ground laser communication system at an AIR laser ground station and the AIRSAT-02 satellite.

Notably, the satellite's hardware remained unchanged. Through in-orbit software reconfiguration, the team fully leveraged the potential of the laser communication payload, increasing its capability from 60 Gbps to 120 Gbps.

During the experiment, the satellite and ground station achieved link acquisition in seconds, with a success rate exceeding 93 percent. The maximum continuous communication duration reached 108 seconds, resulting in a total data acquisition of 12.656 terabits, Li said.

Source(s): Xinhua News Agency

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